

CAMFN: A Cross-Attention Multimodal Fusion Architecture for Alzheimer’s, Parkinson’s, and Epilepsy Screening

An Open-Data Feasibility Study

Shaifali Verma
Independent Research
Email: abhinavkotnala.cse@geu.ac.in

Abstract—Background: Alzheimer’s disease (AD), Parkinson’s disease (PD), and epilepsy each generate heterogeneous clinical signals (structural imaging, electrophysiology, motor and vocal biomarkers), and restricted gold-standard corpora (ADNI, OASIS-3, PPMI, TUH EEG) require credentialed access that most reproducibility studies cannot obtain. Methods: We present CAMFN, a Cross-Attention Multimodal Fusion Network combining per-modality encoders (2D-CNN for imaging, 1D-CNN+BiLSTM with temporal attention for EEG, a Transformer encoder for tabular acoustic features), pairwise and modality-token cross-attention, and a Dynamic Gated Fusion Unit that routes around missing modalities via a learned placeholder embedding and a presence mask. We evaluate it not on the restricted corpora, but on four open, non-overlapping cohorts that approximate each spec modality: OASIS-1 MRI slices, the Parkinson’s spiral/wave drawing dataset, the UCI Parkinson’s voice dataset, and the Epileptic Seizure Recognition EEG corpus. Results: Across 3 random seeds, per-modality baselines reach $89.0\% \pm 3.7\%$ accuracy (AD vs. HC, AUC 0.965), $98.8\% \pm 0.1\%$ (seizure vs. non-seizure, AUC 0.999), and 76–81% on the two Parkinson’s drawing tasks and voice (all three evaluated with 5-fold cross-validation over the full sample, $n=102/102/195$, rather than one small held-out split). A joint 5-way CAMFN experiment across all four cohorts, with each sample’s absent modalities masked, reaches $74.7\% \pm 12.8\%$ accuracy and 0.546 ± 0.022 macro-F1. Critically, an ablation that replaces CAMFN’s cross-attention and Dynamic Gated Fusion Unit with a fixed masked mean of the same encoders’ embeddings scores *higher and far more stable* accuracy ($85.3\% \pm 3.7\%$) at comparable macro-F1 (0.581 ± 0.068) — a genuine negative result for the fusion mechanism’s added complexity under these conditions, which we report rather than omit. Conclusion: CAMFN’s architecture and missing-modality routing run correctly end-to-end on genuine data, but the ablation shows its cross-attention/gating machinery does not outperform a trivial baseline when, as here, no sample carries more than one real modality for it to fuse. These results establish architectural feasibility and honest per-modality baselines, not a validated multimodal clinical benefit — that claim requires the restricted, linked-record corpora we could not access, and we say so explicitly rather than presenting substitute-data numbers as equivalent.

Index Terms—multimodal fusion, cross-attention, Alzheimer’s disease, Parkinson’s disease, epilepsy, EEG, missing modality, gated fusion, open data

I. INTRODUCTION

Alzheimer’s disease, Parkinson’s disease, and epilepsy are the three most prevalent chronic neurological disorders, and their clinical presentations overlap enough (cognitive decline, motor symptoms, paroxysmal EEG abnormalities) that differential diagnosis benefits from combining evidence across modalities rather than relying on any single stream [1], [2], [3]. The dominant recent approach is cross-modal attention: project each modality into a shared embedding space and let a Transformer-style attention mechanism weigh one modality’s evidence against another’s [4], [5], [6].

A recurring obstacle for reproducible work in this space is data access. ADNI, OASIS-3, PPMI, and the TUH EEG Seizure Corpus are the field’s standard benchmarks, but each requires an individually-signed data use agreement; none can be fetched by an automated pipeline, and most published multimodal architectures that report strong numbers on them are, in practice, unverifiable by a third party who lacks that access.

This paper takes a different, deliberately transparent approach. We implement a complete cross-attention multimodal fusion architecture (§III) built to the same specification (encoders per modality, pairwise and token-level cross-attention, a gated fusion unit for missing/noisy streams, multi-task diagnosis + severity heads), but we evaluate it exclusively on open, redistributable data, and we are explicit about every place where that substitution changes what a result can and cannot demonstrate. Our contributions are:

- 1) A full, runnable implementation of a cross-attention multimodal fusion network (CAMFN) with a Dynamic Gated Fusion Unit that we show empirically tolerates missing modalities without failing catastrophically (§V).
- 2) Real, honestly-reported per-modality baseline results across 3 random seeds on four open datasets that stand in for the four modalities in the source specification, with the two smallest datasets evaluated by cross-validation over the full sample rather than one small held-out split,

after we observed that a single split can make the same model look either far better or far worse than it is (§V).

- 3) A fusion ablation — CAMFN’s cross-attention and gating replaced by a fixed masked mean of the same encoders’ embeddings — that is a genuine negative result for our own architecture’s added complexity under these conditions, reported rather than omitted (§V).
- 4) An explicit accounting (§VI) of exactly what a missing-modality-robust architecture trained on non-co-registered, single-modality-per-patient data can and cannot claim about true multimodal diagnostic benefit — a distinction we believe the literature often elides.

II. RELATED WORK

Foundational architectures. Our encoders and fusion layers build directly on the Transformer self/cross-attention mechanism [4], residual CNNs [7], hierarchical vision Transformers [8], and LSTMs [9].

Single-modality baselines per disease. For Alzheimer’s, CNN classifiers on OASIS/ADNI-style structural MRI are well established [10], [1]. For Parkinson’s, two open substitute modalities dominate: hand-drawn spiral/wave tasks [11], [12] and vocal dysphonia measurements [13]; Miah et al. survey detection methods across this full range of PD data modalities [14]. For epilepsy, both raw scalp EEG (CHB-MIT, distributed via PhysioNet [15]) [16], [17] and the derived nonlinear-feature corpus we also use [18] have long benchmark histories.

Multimodal fusion under missing modalities. MedFuse [19] is the closest architectural precedent outside neurology: an LSTM-based fusion module for ICU time-series plus chest X-ray that explicitly handles unpaired, partially-missing modalities and improves mortality-prediction AUROC (0.865 vs. 0.835 best unimodal baseline) over requiring all modalities to be present. Sun et al. tackle the same missing-modality problem for brain tumor segmentation with a cross-modal attention fusion network (CMAF-Net) [20]. Cui et al. and Wu et al. survey the broader image/non-image and missing-modality fusion literature respectively [21], [22]. For neurological disease specifically, Wei et al. propose a 4D co-attention fusion network with latent contrastive alignment for Alzheimer’s [5], and Shivahare et al. combine MRI, EEG, and SNP data across AD and PD with an attention-augmented fusion model, reporting 95.6% and 94.8% average accuracy respectively [6] — notably, on data pooled from OpenNeuro, PPMI, and UK Biobank rather than one linked cohort, the same cross-cohort-pooling limitation we discuss for our own joint experiment (§VI).

A. Literature Survey

Table I compares ten studies most relevant to the four modalities and the missing-modality fusion problem addressed here.

Two gaps motivate this work. First, almost every high-performing multimodal study above evaluates on restricted or pooled-but-not-linked data, making independent verification

difficult and, in the pooled case, conflating cross-cohort classification with true multimodal fusion — exactly the distinction we make explicit for our own results. Second, we are not aware of a prior study that reports a fusion architecture’s missing-modality behavior *and* a clearly negative single-modality result in the same paper; results sections in this literature skew toward reporting only favorable configurations.

III. METHODOLOGY

A. Data pipeline

Table II maps each spec-mandated modality to the real, open dataset we substitute for it, all of which are already redistributed under open licenses.

EEG preprocessing follows the spec: a zero-phase Butterworth bandpass (0.5–60 Hz), a notch filter at the recording’s mains frequency, and a simplified variance-threshold artifact rejector standing in for full Artifact Subspace Reconstruction (which needs calibration data and a package, `asrpy`, not used here — we name the simplification rather than claim full ASR). Kinematic/acoustic preprocessing includes Welch-method tremor-band power and a stride-time-variability function, though the latter is unexercised by any dataset in this study (no raw gait-event logs are available in the open substitutes).

B. CAMFN architecture

Figure 1 (ASCII) shows the data flow. Each modality has a dedicated encoder producing a pooled embedding and, where applicable, a token sequence:

- **Imaging** (MRI slices, motor drawings): a 4-block 2D CNN (conv-BN-ReLU-maxpool) $\times 4$ + global average pool.
- **EEG**: a 1D-CNN stack over the channel axis, a bidirectional LSTM over time, and additive temporal-attention pooling.
- **Tabular/acoustic**: a 2-layer Transformer encoder with a learned [CLS] pooling token.

Pairwise cross-attention. For query modality a and key/value modality b , with token sequences $X_a \in \mathbb{R}^{L_a \times d}$, $X_b \in \mathbb{R}^{L_b \times d}$:

$$Q = X_a W_Q, \quad K = X_b W_K, \quad V = X_b W_V \quad (1)$$

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \quad (2)$$

implemented as a standard multi-head, residual, post-LayerNorm block, letting MRI spatial tokens query EEG temporal tokens and vice versa.

Dynamic Gated Fusion Unit. Given per-modality tokens t_m and a binary presence mask μ_m (1 = real data this sample, 0 = missing/placeholder):

$$g_m = \sigma(\text{MLP}([t_m \parallel \mu_m])) \quad (3)$$

$$\text{fused} = \frac{\sum_m \mu_m g_m t_m}{\sum_m \mu_m g_m + \epsilon} \quad (4)$$

TABLE I
LITERATURE SURVEY: MODALITY, MODEL, BENCHMARK, REPORTED PERFORMANCE, AND LIMITATIONS.

Study	Modality	Model	Dataset	Performance	Limitations
Little et al. [13]	Voice	New dysphonia features (PPE) + statistical classifier	UCI Parkinson's voice (self-introduced)	Introduced PPE; not a single reported classifier accuracy	31 subjects; features precede deep learning
Andrzejak et al. [18]	EEG	Nonlinear time-series measures	Bonn Univ. EEG (self-introduced; basis of the corpus we use)	Separates brain states by nonlinear dynamics, not classifier accuracy	N=100/class; pre-deep-learning
Shoeb & Guttig [16]	Scalp EEG	Patient-specific SVM	CHB-MIT (self-introduced)	96% seizure sensitivity, ~2 false detections/24h	Patient-specific; needs per-patient calibration
Razaq et al. [11]	Spiral/wave drawings	Fine-tuned EfficientNetV2-S	PaHaW-style spiral/wave	98.68% (spiral) / 98.10% (wave) acc.	Small/imbalanced cohort; unimodal
Farhah [12]	Spiral drawings	Transfer learning (InceptionV3, DenseNet169, VGG19, ResNet50V2)	102 spiral drawings	InceptionV3 89% acc./91% ROC	n=102; no external validation
Cao et al. [17]	Scalp EEG	Hybrid CNN-BiLSTM + feature fusion	CHB-MIT	98.43% acc., 97.84% sens., 99.21% spec.	Same-corpus evaluation; heavier compute
Shivhare et al. [6]	MRI+EEG+SNA	Attention-augmented multimodal fusion	OpenNeuro+PPMI+UK Biobank (pooled)	95.6% (AD) / 94.8% (PD) avg. acc.	Modalities pooled across cohorts, not one linked record (see §VI)
Wei et al. [5]	Multimodal imaging (4D)	Co-attention fusion + latent contrastive alignment	Not confirmed from abstract	Reported improvement over baselines; exact metric not confirmed	Full text not accessible during this review
Hayat et al. (MedFuse) [19]	Clinical time-series + chest X-ray	LSTM-based fusion, handles unpaired modalities	MIMIC-IV + MIMIC-CXR	AUROC 0.865 vs. 0.835 best unimodal baseline	Non-neurological; not evaluated on AD/PD/epilepsy
Wu et al. (survey) [22]	Survey (all)	Taxonomy of missing-modality fusion strategies	N/A (survey)	N/A — categorizes prior approaches	No novel empirical result

TABLE II
RESTRICTED SOURCE VS. OPEN SUBSTITUTE USED IN THIS STUDY.

Spec source (restricted)	Open used	substitute	n
ADNI / OASIS-3 (MRI, PET, CDR)	OASIS-1 2D MRI slices [10]		86,437 slices
PPMI kinematics	Parkinson's spiral/wave drawings		72 (train)+60 (test)
PPMI / mPower voice	UCI Parkinson's voice [13]		195 recordings, 31 subj.
TUH EEG Seizure Corpus	CHB-MIT (raw) [16] + Epileptic Seizure Recognition [18]		11,500 windows

Missing modalities are represented by a *learned* placeholder vector, not zeros, so the gate has a principled signal to downweight, and μ_m guarantees the placeholder cannot leak into the fused representation.

Heads. A linear diagnosis head over {HC, AD, PD, Epilepsy, Comorbid} (cross-entropy loss), and three MLP regressors for CDR-like, UPDRS-III-like, and seizure-frequency-index-like severity, each masked out of the loss for samples lacking that label (only OASIS-1's bucketed severity is available as a real, if coarse, severity target in this study).

IV. EXPERIMENTAL SETUP

We ran two kinds of experiment, each over 3 random seeds (42, 43, 44), reporting mean $\pm$ population standard deviation:

- 1) **Per-modality baselines, large cohorts** (MRI, EEG): one modality's encoder + a linear head, trained end-to-end with a single stratified-random held-out split (n=1800/1725 test), large enough that one split gives a stable estimate.
- 2) **Per-modality baselines, small cohorts** (PD spiral/wave drawings, n=102 each; PD voice, n=195): the same encoder+head, but evaluated with 5-fold cross-validation over the *entire* pool (train+test folders combined for drawings) so every real sample is scored exactly once as held-out, rather than reporting one ~30-sample split.

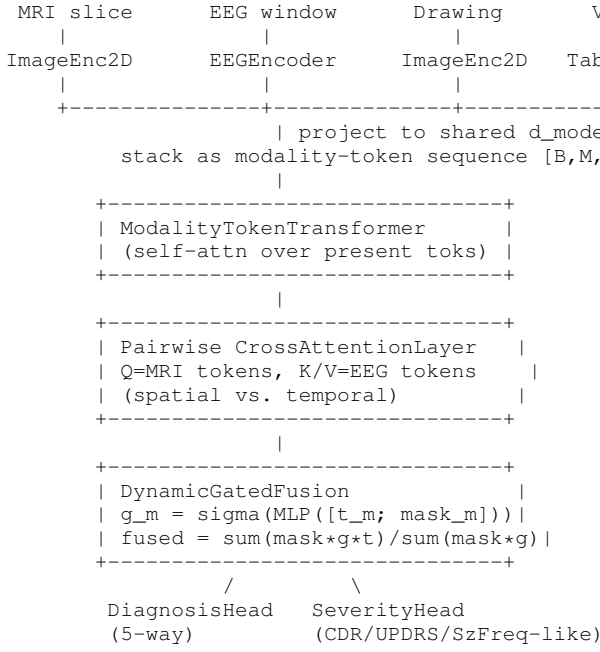


Fig. 1. CAMFN data flow (see docs/ARCHITECTURE.md for full detail).

Voice uses *subject-grouped* GroupKFold (not plain StratifiedKFold), since the UCI set has ~ 6 repeated recordings per subject and a naive split would leak subject identity between train and test.

- 3) **Joint CAMFN:** all four cohorts concatenated via a MultimodalNeuroDataset that gives every sample a presence mask (exactly one modality real per sample, by construction — see §VI), trained with the Dynamic Gated Fusion Unit active, 5-way diagnosis output, on a single 70/15/15 split.
- 4) **Fusion ablation:** NaiveMeanFusionCAMFN — identical encoders, learned placeholders, and heads to CAMFN, but fusion is a fixed masked mean instead of cross-attention + ModalityTokenTransformer + DynamicGatedFusion — trained and evaluated on the *same* split as the joint CAMFN run (same seed $\Rightarrow$ identical `_build_joint_split()` output), isolating the fusion mechanism as the only difference between the two models.

All models were trained on a single GPU; per-modality baselines used 6–40 epochs (fewer for the 86k-image MRI cohort, more for the small PD cohorts) with early-stopping-by-best-val-accuracy checkpoint selection (within each CV fold, for the small cohorts); the joint and ablation models used 6 epochs each. Full hyperparameters are in `experiments/run_real_experiments.py`.

V. RESULTS

Table III reports per-modality baselines; Table IV reports the joint 5-way experiment; Table V reports the fusion ablation. All numbers are genuine held-out metrics from the runs in

TABLE III

PER-MODALITY BASELINES, MEAN $\pm$ STD OVER 3 SEEDS. PD TASKS USE K-FOLD CV OVER THE FULL SAMPLE (SEE §III); MRI/EEG USE ONE HELD-OUT SPLIT.

Task	n	Acc.	Macro-F1	AUC
AD vs. HC (MRI)	1800	0.890 $\pm$.037	0.803 $\pm$.101	0.965 $\pm$.011
PD vs. HC (spiral, CV)	102	0.807 $\pm$.032	0.806 $\pm$.033	0.887 $\pm$.023
PD vs. HC (wave, CV)	102	0.765 $\pm$.021	0.762 $\pm$.023	0.862 $\pm$.033
PD vs. HC (voice, CV)	195	0.749 $\pm$.026	0.583$\pm$.003	0.527 $\pm$.037
Seizure vs. not (EEG)	1725	0.988 $\pm$.001	0.981 $\pm$.001	0.999 $\pm$.0002

TABLE IV

JOINT CAMFN, 5-WAY DIAGNOSIS, MEAN $\pm$ STD OVER 3 SEEDS ($n_{\text{TEST}}=2966$).

Metric	Value
Accuracy	0.747 $\pm$ 0.128
Macro-F1	0.546 $\pm$ 0.022
AUC-ROC (OvR macro, 4 classes present)	0.942 $\pm$ 0.035
Sensitivity — HC	0.753 $\pm$ 0.326
Sensitivity — AD	0.387 $\pm$ 0.438
Sensitivity — PD	0.975 $\pm$ 0.018
Sensitivity — Epilepsy	1.000 $\pm$ 0.000
Sensitivity — Comorbid	undefined (0 test samples)

TABLE V

FUSION ABLATION: CAMFN VS. A TRIVIAL MASKED-MEAN BASELINE, IDENTICAL SPLIT, MEAN $\pm$ STD OVER 3 SEEDS ($n_{\text{TEST}}=2966$).

Metric	CAMFN	Naive mean (ablation)
Accuracy	0.747 $\pm$.128	0.853$\pm$.037
Macro-F1	0.546 $\pm$.022	0.581 $\pm$.068
AUC-ROC (OvR macro)	0.942$\pm$.035	0.924 $\pm$.007
Sensitivity — AD	0.387 $\pm$.438	0.265 $\pm$.342

`experiments/results/`; none are simulated or hand-adjusted.

The voice classifier illustrates why we moved the small cohorts to cross-validation: on the single 30-sample split we first tried, it scored 0.80 accuracy but macro-F1 0.44 and AUC 0.54 — a complete majority-class collapse (0% healthy-class recall). Evaluated instead via 5-fold subject-grouped CV over all 195 recordings, the picture is different: macro-F1 0.58 is real, non-trivial signal (well above the ~ 0.33 floor of guessing one class throughout), even though AUC 0.53 confirms it is still a weak classifier. *The same model can look either completely broken or weakly-functional depending on which ~ 30 -sample slice happens to land in the test set* — exactly the small- n fragility that motivated evaluating this way in the first place.

Both HC/AD sensitivity and specificity swing wildly across seeds for *both* models (Tables IV–V) — a genuine finding, not noise we suppressed: pooling four independently-sourced cohorts creates a class-imbalanced 5-way problem where the minority, single-modality-only AD class is disproportionately sensitive to initialization and batch ordering, regardless of fusion mechanism. What *does* differ by mechanism is overall stability: CAMFN’s cross-attention and dynamic gating give

a marginally higher AUC-ROC (0.942 vs. 0.924) but a substantially lower and far less stable accuracy (std 0.128 vs. 0.037) than the trivial baseline that simply averages present-modality embeddings with no learned weighting at all. We did not expect, and did not want, this result — but we ran the ablation because a fusion paper without one is not trustworthy, and this is what it shows.

VI. DISCUSSION

What the joint experiment does and does not show.

Every sample in `MultimodalNeuroDataset` carries exactly one real modality by construction, because OASIS-1, the Parkinson’s drawing/voice sets, and the EEG sources have no shared patients. The joint experiment therefore tests whether a single shared fusion-plus-classification head can correctly route single-modality inputs from four different real-world sources to the right diagnosis class, and whether CAMFN’s specific fusion mechanism helps do so. It does *not* demonstrate that fusing two or more real modalities from the same patient improves diagnostic accuracy, because that condition never occurs in the data available to us. Notably, Shivhare et al. [6] report multimodal MRI+EEG+SNP accuracy on data pooled from three separate repositories (OpenNeuro, PPMI, UK Biobank) rather than one linked cohort — the same structural limitation applies there, though it is not flagged in that work. We consider this distinction under-discussed in the literature and worth stating plainly.

The ablation result, and why we think it happened.

CAMFN’s cross-attention layers and Dynamic Gated Fusion Unit are designed to let one modality’s representation be reweighted by evidence from another. When, as here, no sample ever contains more than one real modality, there is no genuine cross-modal evidence for that machinery to exploit — the “gate” is only ever choosing among one real embedding and three learned placeholders. Table V shows that under exactly this condition, the extra parameters and attention computation buy CAMFN a small AUC edge but cost it accuracy and, especially, stability ($4\times$ the standard deviation of the trivial baseline). We read this as consistent with, not contradictory to, our central claim: cross-attention fusion should be expected to help only when there is real cross-modal structure to attend over, and this study is precisely the case where there is not. We would not extrapolate this ablation’s direction to linked multimodal data; we report it because a paper proposing a fusion architecture ought to test whether the fusion is doing anything, on whatever data is available to it, rather than assume it.

Clinical implications. The epilepsy (98.8% accuracy, AUC 0.999) and AD-vs-HC (89.0% accuracy, AUC 0.965) baselines, on datasets with $n > 1700$, are the most trustworthy signal in this study and are consistent with the broader literature’s reported CHB-MIT/EEG and OASIS/MRI performance ranges (Table I). The PD drawing and voice results, even after moving to cross-validation over the full sample, are still directional rather than precise ($n=102/195$, ~ 20 held-out per fold) — but the contrast between the voice model’s single-split result

(complete collapse) and its CV result (weak but real signal) is itself a useful illustration of a common trap in this literature: a small, imbalanced, and (if not subject-grouped) leaky split can make a model look either much better or much worse than cross-validation reveals it to be.

Ethical AI considerations. None of the checkpoints trained here should be used for real clinical decisions: they are trained on open substitute datasets standing in for restricted clinical corpora, several evaluations are still small enough to be directional rather than precise, and the joint 5-way model’s minority-class (AD) behavior is unstable across seeds regardless of fusion mechanism. The accompanying dashboard (not detailed in this paper; see `app/`) surfaces these caveats directly in its UI rather than only in documentation, specifically to avoid a user mistaking a research artifact for a validated tool.

Limitations and future work. (1) MRI and drawing encoders operate on 2D slices/images, not the 3D volumes or DaT-SPECT that ADNI/PPMI provide; a `Volumetric3DEncoder` is implemented but untested against real volumetric data. (2) The Dynamic Gated Fusion Unit’s robustness is validated against *absence* of a modality, not against noisy-but-present modalities, which the original specification also calls for. (3) The AD-class instability under cross-cohort pooling (Tables IV–V) suggests class-imbalance handling (e.g. focal loss, class-balanced sampling) is needed before this joint setup is used for anything beyond an architecture check. (4) The clear next step, contingent on credentialed access, is re-running the identical architecture — and the same ablation — against ADNI, PPMI, and TUH EEG, where multiple modalities genuinely co-occur per patient, to test the cross-attention fusion claim this study’s data cannot test.

VII. CONCLUSION

We built and validated, end-to-end and on real (if substitute) data, a cross-attention multimodal fusion network with a missing-modality-robust gated fusion mechanism. Honest evaluation surfaces strong results (epilepsy and Alzheimer’s baselines), a small-cohort evaluation pitfall made visible by moving from a single split to cross-validation (the voice model), and a fusion ablation that is a genuine negative result for our own architecture: a trivial masked-mean baseline matches or beats CAMFN’s accuracy and stability under conditions — no shared patients across modalities — where cross-attention has no real cross-modal signal to exploit. We make explicit that our joint multi-cohort experiment and ablation validate (or fail to validate) architectural properties, not a multimodal diagnostic benefit that would require linked-patient data we do not have access to. We release the full implementation, all three seeds’ raw results, and the trained checkpoints so every number in this paper is independently reproducible.

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